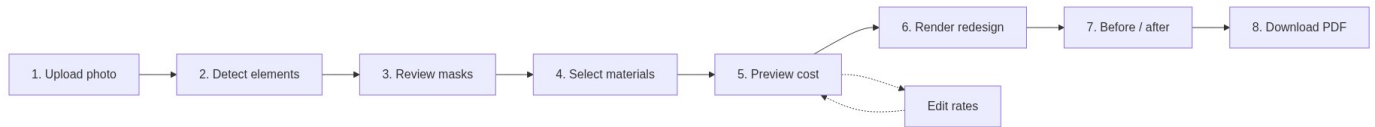


03 — User Workflow

Problem-statement deliverable: *A user workflow*

1. End-to-end journey



2. Step-by-step

Step 1 — Upload (/)

- User drops or selects a JPEG / PNG / WebP of the house exterior.
- Backend **Ingestion Engine** validates type/size, EXIF-oriens, resizes long edge, and may return **warnings** (blurry, too dark, too bright).
- User is redirected to `/studio/{projectId}`.

Tip for better results: Straight-on facade photo in good daylight; avoid extreme zoom or heavy obstruction.

Step 2 — Detect elements (Studio)

- User clicks **Detect Elements**.
- Default categories: wall, balcony, rooftop, gate, window, door (railing / pillar available in taxonomy).
- Grounding DINO proposes boxes from text prompts; SAM produces masks.
- **Walls** are derived as building silhouette minus openings (more reliable than direct “wall” detection).
- Colored overlays appear on the canvas.

Step 3 — Review / refine masks

- User selects a category and can **brush / erase** the mask (`MaskEditor`) so missed areas or false positives are corrected.
- Changes are saved to the RegionMap via `PUT /api/segmentation/{id}/regions/{category}`.
- This human-in-the-loop step turns an imperfect model into a usable plan.

Step 4 — Select materials

- For each detected category, user picks:
- **Solid paint** + color, or
- A **texture** (wall cladding, wall tiles, texture patterns) from the catalog.

- Selections can be changed and re-rendered.

Step 5 — Preview cost

- Studio shows a live **cost estimate** panel.
- User sets approximate **facade width × height (m)** (defaults 12 m × 9 m).
- Quantities use mask pixel share × facade area × default 10% waste (area materials).
- Gates / doors / windows / pillars price by **piece count**.
- Optional: open **/cost** to edit material rates (INR); estimates recalculate from stored rates.

Step 6 — Render redesign

- User clicks **Apply finishes & render**.
- Classical rendering applies paint (shade-preserving recolor) or tiles textures inside masks, in a fixed layer order (walls first, openings later).
- Output preserves original geometry and approximate lighting.

Step 7 — Before / after

- Side-by-side or slider comparison of original vs redesigned image.
- User may change materials and re-render.

Step 8 — Download report

- After a successful render, user downloads a **PDF** containing:
- Branding / title
- Original and redesigned images
- Selected materials
- Quantity / cost breakdown (INR)
- Advisory disclaimer

The PDF is intended as a **discussion document** with contractors, not a formal quotation.

3. Screens and responsibilities

Screen	Responsibility
/	Capture project photo

/studio/[projectId]	Detect, edit, assign finishes, estimate, render, export
/cost	Maintain editable rate card shared across projects

4. Typical timing (local CPU)

Stage	Expected order of magnitude
Upload + ingest	under 2 s
First segmentation (cold models)	tens of seconds to a few minutes
Later segmentation / same session	shorter (models warm)
Cost estimate	under 1 s
Classical render	a few seconds
PDF	a few seconds

Exact times depend on CPU, RAM, and image size.